

Q1 JEE Main 2020 - 2 September (Morning)

Two identical strings X and Z made of same material have tension T_x and T_z in them. If their fundamental frequencies are 450Hz and 300Hz, respectively, then the ratio T_x/T_z is

- (A) 1.5
- (B) 2.25
- (C) 0.44
- (D) 1.25

Q2 JEE Main 2020 - 2 September (Morning)

Train A and train B are running on parallel tracks in the opposite directions with speeds of 36 km/hour and 72 km/hour, respectively. A person is walking in train A in the direction opposite to its motion with a speed of 1.8km/ hour. Speed (in ms^{-1}) of this person as observed from train B will be close to : (take the distance between the tracks as negligible)

- (A) 30.5ms^{-1}
- (B) 29.5ms^{-1}
- (C) 31.5ms^{-1}
- (D) 28.5ms^{-1}

Q3 JEE Main 2020 - 2 September (Evening)

A wire of density $9 \times 10^{-3} \text{kgcm}^{-3}$ is stretched between two clamps 1m apart. The resulting strain in the wire is 4.9×10^{-4} . The lowest frequency of the transverse vibrations in the wire is (Young's modulus of wire $Y = 9 \times 10^{10} \text{Nm}^{-2}$), (to the nearest integer),

Q4 JEE Main 2020 - 3 September (Morning)

A uniform thin rope of length 12m and mass 6kg hangs vertically from a rigid support and a block of mass 2kg is attached to its free end. A transverse short wavetrain of wavelength 6cm is produced at the lower end of the rope. What is the wavelength of the wavetrain (in cm) when it reaches the top of the rope?

- (A) 6
- (B) 3
- (C) 12

(D) 9

Q5 JEE Main 2020 - 4 September (Morning)

For a transverse wave travelling along a straight line, the distance between two peaks (crests) is 5m, while the distance between one crest and one trough is 1.5m. The possible wavelengths (in m) of the waves are

(A) 1, 2, 3, ...

(B) 1, 3, 5, ...

(C) $\frac{1}{1}, \frac{1}{3}, \frac{1}{5}, \dots$ (D) $\frac{1}{2}, \frac{1}{4}, \frac{1}{6}, \dots$ **Q6 JEE Main 2020 - 4 September (Evening)**

The driver of a bus approaching a big wall notices that the frequency of his bus's horn changes from 420Hz to 490Hz when he hears it after it gets reflected from the wall. Find the speed of the bus if speed of the sound is 330ms^{-1}

(A) 61kmh^{-1} (B) 91kmh^{-1} (C) 81kmh^{-1} (D) 71kmh^{-1} **Q7 JEE Main 2020 - 5 September (Morning)**

Assume that the displacement (s) of air is proportional to the pressure difference (Δp) created by a sound wave. Displacement (s) further depends on the speed of sound (v), density of air (ρ) and the frequency (f). If $\Delta p \sim 10\text{Pa}$, $v \sim 300\text{m/s}$, $\rho \sim 1\text{kg/m}^3$ and $f \sim 1000\text{Hz}$ then s will be of the order of (take the multiplicative constant to be 1)

(A) $\frac{3}{100}\text{mm}$

(B) 10mm

(C) 1mm

(D) $\frac{1}{10}\text{mm}$ **Q8 JEE Main 2020 - 5 September (Morning)**

In a resonance tube experiment when the tube is filled with water up to a height of 17.0cm from bottom, it resonates with a given tuning fork. When the water level is raised the next resonance with the same tuning fork occurs at a height of 24.5cm. If the velocity of sound in air is 330m/s, the tuning fork frequency is

- (A) 2200Hz
- (B) 3300Hz
- (C) 1100Hz
- (D) 550Hz

Q9 JEE Main 2020 - 5 September (Evening)

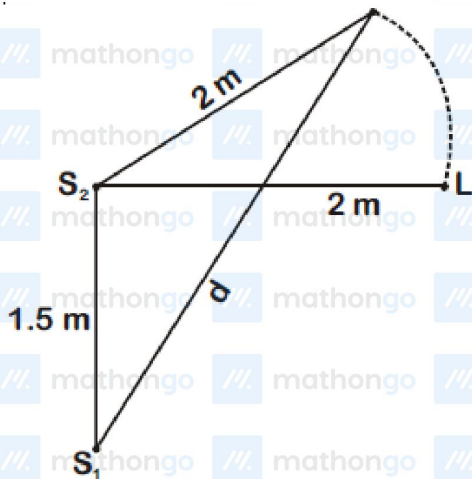
A driver in a car, approaching a vertical wall notices that the frequency of his car horn, has changed from 440Hz to 480Hz, when it gets reflected from the wall. If the speed of sound in air is 345m/s, then the speed of the car is

- (A) 36km/hr
- (B) 54km/hr
- (C) 24km/hr
- (D) 18km/hr

Q10 JEE Main 2020 - 5 September (Evening)

Two coherent sources of sound, S_1 and S_2 , produce sound waves of the same wavelength, $\lambda = 1\text{m}$, in phase. S_1 and S_2 are placed 1.5m apart (see fig). A listener, located at L, directly in front of S_2 finds that the intensity is at a minimum when he is 2m away from S_2 . The listener moves away from S_1 , keeping his distance from S_2 fixed. The adjacent maximum of intensity is observed when the listener is

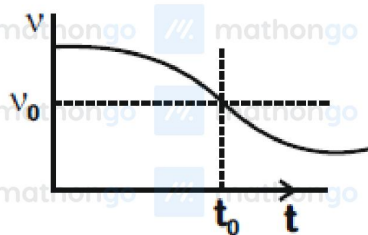
at a distance d from S_1 . Then, d is



- (A) 5m
- (B) 2m
- (C) 12m
- (D) 3m

Q11 JEE Main 2020 - 6 September (Morning)

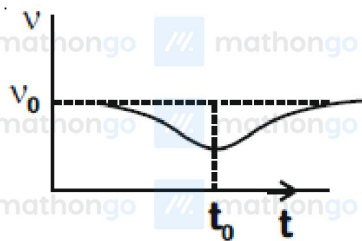
A sound source S is moving along a straight track with speed v , and is emitting sound of frequency ν_0 (see figure). An observer is standing at a finite distance, at the point O , from the track. The time variation of frequency heard by the observer is best represented by (t_0 represents the instant when the distance between the source and observer is minimum)



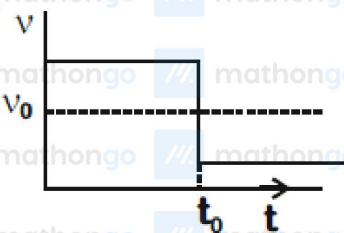
- (A)



(B)



(C)



(D)

Q12 JEE Main 2020 - 7 January (Morning)

Speed of a transverse wave on a straight wire (mass $6g$, length $60cm$ and area of cross-section $1.0mm^2$) is $90ms^{-1}$. If the Young's modulus of wire is $16 \times 10^{11}Nm^{-2}$, the extension of wire over its natural length is :

- (A) 0.03 mm
- (B) 0.02 mm
- (C) 0.01 mm
- (D) 0.04 mm

Q13 JEE Main 2020 - 7 January (Evening)

A stationary observer receives sound from two identical tuning forks, one of which approaches and the other one recedes with the same speed (much less than the speed of sound). The observer hears

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2beats/sec . The oscillation frequency of each tuning fork is $\nu_o = 1400\text{ Hz}$ and the velocity of sound in air is 350 m/s . The speed of each tuning fork is close to :

- (A) $\frac{1}{4}$
- (B) 4
- (C) 2
- (D) $\frac{1}{2}$

Q14 JEE Main 2020 - 8 January (Morning)

A *one* metre long (both ends open) organ pipe is kept in a gas that has double the density of air at *STP*. Assuming the speed of sound in air at *STP* is 300 m/s , the frequency difference between the fundamental and second harmonic of this pipe is --- Hz

Q15 JEE Main 2020 - 8 January (Evening)

A transverse wave travels on a taut steel wire with a velocity of v when tension in it is $2.06 \times 10^4\text{ N}$. When the tension is changed to T , the velocity changed to $v/2$. The value of T is close to :

- (A) $5.15 \times 10^3\text{ N}$
- (B) $8.24 \times 10^4\text{ N}$
- (C) 6×10^4
- (D) $5.15 \times 10^4\text{ N}$

Q16 JEE Main 2020 - 9 January (Morning)

Three harmonic waves having equal frequency ν and same intensity I_o , have phase angles $0, \frac{\pi}{4}$ and $-\frac{\pi}{4}$ respectively. When they are superimposed the intensity of the resultant wave is close to :

- (A) I_o
- (B) 0
- (C) $5.8I_o$
- (D) $0.2 I_o$

Q17 JEE Main 2020 - 9 January (Evening)

A wire of length L and mass per unit length $6.0 \times 10^{-3}\text{ kgm}^{-1}$ is put under tension of 540 N . Two consecutive frequencies that it resonates at are 420 Hz and 490 Hz . Then L in meters is :

Wave Mechanics

Questions with Answer Keys

JEE Main 2020 Chapterwise

Physics

(A) 8.1 m

(B) 2.1 m

(C) 1.1 m

(D) 5.1 m

Wave Mechanics

Questions with Answer Keys

JEE Main 2020 Chapterwise

Physics

Answer Key

Q1 (B)

Q2 (B)

Q3 (35)

Q4 (C)

Q5 (C)

Q6 (B)

Q7 (A)

Q8 (A)

Q9 (B)

Q10 (D)

Q11 (A)

Q12 (A)

Q13 (A)

Q14 (105.75)

Q15 (A)

Q16 (C)

Q17 (B)

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Wave Mechanics

Hints & Solutions

JEE Main 2020 Chapterwise

Physics

Q1

$$f = \frac{V}{2L}$$
$$\Rightarrow \frac{450}{300} = \sqrt{\frac{T_x}{T_z}}$$
$$\Rightarrow \frac{T_x}{T_z} = 2.25$$

Q2



$$V_P = (10 - 0.5)$$
$$= 9.5 \text{ m/s}$$

$$V_{PB} = 29.5 \text{ m/s}$$

Q3

$$m = A l \sigma \quad \therefore \mu = \left(\frac{m}{l} \right) = A \sigma$$

$$\text{Now, } v = \sqrt{\frac{T}{\mu}} \text{ and } \frac{T}{A} = Y (\text{strain})$$

$$\Rightarrow v = \sqrt{\frac{T}{\sigma A}} \Rightarrow \frac{T}{A} = 9 \times 10^{10} \times 4.9 \times 10^{-4}$$

$$\therefore v = \sqrt{\frac{9 \times 10^{10} \times 4.9 \times 10^{-4}}{9 \times 10^3}}$$

$$\Rightarrow v = 70 \text{ m/s}$$

$$\therefore f_0 = \frac{v}{2L} = \frac{70}{2 \times 1} = 35 \text{ Hz}$$

Q4

Wave Mechanics

Hints & Solutions

JEE Main 2020 Chapterwise

Physics

$$\begin{aligned}f &= \frac{v}{\lambda} \\ \frac{V_1}{\lambda_1} &= \frac{V_2}{\lambda_2} \\ \Rightarrow \lambda_2 &= \lambda_1 \frac{V_2}{V_1} \\ &= \lambda_1 \sqrt{\frac{T_2}{T_1}} = \lambda_1 \sqrt{\frac{8g}{2g}} \\ \Rightarrow \lambda_2 &= 12\text{cm}\end{aligned}$$

Q5

$$\left(\frac{\lambda}{2}\right) \times n = 1.5, \quad n \text{ is odd integer}$$

$$\lambda \times m = 5, \quad m \text{ is integer}$$

$$\Rightarrow \lambda \text{ can be } \frac{1}{1}, \frac{1}{3}, \frac{1}{5}$$

$$\text{as } \lambda = \frac{3}{n} \text{ and } \lambda = \frac{5}{m}$$

Q6

$$\frac{330+v}{330-v} \times 420 = 490$$

$$\Rightarrow \frac{330+v}{330-v} = \frac{7}{6}$$

$$\Rightarrow 13v = 330$$

$$\Rightarrow v = \frac{330}{13} = 25.38\text{m/s}$$

$$= 91\text{km/h}$$

Q7

$$\therefore \Delta P_0 = \frac{B\omega}{v} \times S_0$$

$$\Rightarrow S = \frac{\Delta P \times v}{B\omega} = \frac{\Delta P \times v}{\rho v^2 \times 2\pi v}$$

$$\Rightarrow S \propto \frac{\Delta P}{\rho v f}$$

$$\Rightarrow S = \frac{\Delta P}{\rho v f}$$

$$= \frac{10}{1 \times 300 \times 1000}$$

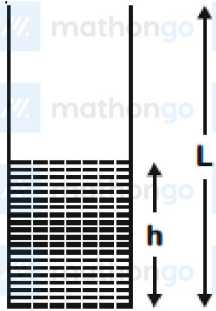
$$= \frac{1}{30} \text{mm}$$

$$\approx \frac{3}{100} \text{mm}$$

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Q8



$$L - h_1 = n \frac{\lambda}{2} + \frac{\lambda}{4}$$

$$L - h_2 = (n - 1) \frac{\lambda}{2} + \frac{\lambda}{4}$$

$$\Rightarrow \frac{\lambda}{2} = h_2 - h_1 = 24.5 - 17.0 = 7.5 \text{ cm}$$

$$\Rightarrow \lambda = 15 \text{ cm}$$

$$u = f\lambda$$

$$f = 2200 \text{ Hz}$$

Q9

$$\frac{345+u}{345-v} \times 440 = 480$$

$$\Rightarrow 11 \times 345 + 11v = 12 \times 345 - 12v$$

$$\Rightarrow v = \frac{345}{23} = 15 \text{ m/s}$$

$$= 54 \text{ km/h}$$

Q10

$$\text{At } L, S_1 L - S_2 L = 2.5 - 2.0 = 0.5 \text{ m}$$

$$= \frac{\lambda}{2}$$

For adjacent maximum

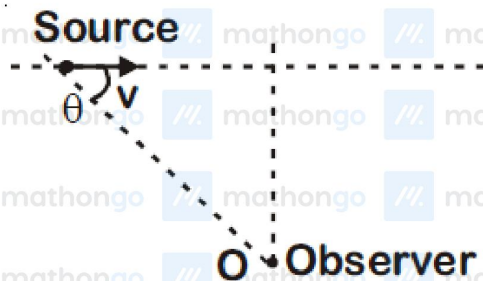
$$S_1 L - S_2 L = \lambda$$

$$S_1 L = \lambda + S_2 L$$

$$= 1 + 2$$

$$= 3 \text{ m}$$

Q11



While approaching

$$v = v_0 \left(\frac{c}{c - v \cos \theta} \right)$$

While receding

$$v = v_0 \left(\frac{c}{c + v \cos \theta} \right)$$

Q12

$$v = \sqrt{\frac{T}{\mu}}$$

$$T = \mu v^2$$

$$\frac{\mu v^2}{A} = Y \frac{\Delta \ell}{\ell}$$

$$\Delta \ell = \frac{\mu v^2 \ell}{AY}$$

after substituting value of μ , v , ℓ , A and Y we get

$$\Delta \ell = 0.03 \text{ mm}$$

Q13

$$f_0 \left(\frac{C}{C - V} \right) - f_0 \left(\frac{C}{C + V} \right) = 2$$


$V \xrightarrow{\quad} S_1 \quad \text{observer} \quad S_2 \xleftarrow{\quad} V$

$$V = \frac{1}{4} m/s$$

Q14

$$V = \sqrt{\frac{B}{\rho}}$$

$$\frac{V_{\text{pipe}}}{V_{\text{air}}} = \frac{\sqrt{\frac{B}{2\rho}}}{\sqrt{\frac{B}{\rho}}} = \frac{1}{\sqrt{2}}$$

$$V_{\text{pipe}} = \frac{V_{\text{air}}}{\sqrt{2}}$$

$$f_n = \frac{(n+1)V_{\text{pipe}}}{2\ell}$$

$$f_1 - f_0 = \frac{V_{\text{pipe}}}{2\ell} = \frac{300}{2\sqrt{2}}$$

$$= 105.75 \text{ Hz (If } \sqrt{2} = 1.41)$$

$$= 106.05 \text{ Hz (If } \sqrt{2} = 1.414)$$

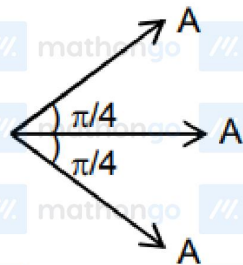
Q15

$$v \propto \sqrt{T}$$

$$\Rightarrow \frac{v_1}{v_2} = \sqrt{\frac{T_1}{T_2}} \Rightarrow \frac{v}{(v/2)} = \sqrt{\frac{2.06 \times 10^4}{T}}$$

$$\Rightarrow T = \frac{2.06 \times 10^4}{4} N = 0.515 \times 10^4 N$$

Q16



$$A_{res} = (\sqrt{2} + 1) A$$

$$I_{res} = (\sqrt{2} + 1)^2 I_0$$

$$= (3 + 2\sqrt{2}) I_0 = 5.8 I_0$$

Q17

Fundamental frequency = $490 - 420 = 70 \text{ Hz}$

$$70 = \frac{1}{2\ell} \sqrt{\frac{T}{\mu}}$$

$$\Rightarrow 70 = \frac{1}{2\ell} \sqrt{\frac{540}{6 \times 10^{-3}}}$$

$$\Rightarrow \ell = \frac{1}{2 \times 70} \sqrt{90 \times 10^3} = \frac{300}{140}$$

$$= \ell \approx 2.14 \text{ m}$$